STALKS OF F_! FOR SEPARATED LOCALLY QUASI-FINITE MORPHISMS AND EXACTNESS

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Allowed standard prerequisites (editorial).

- Exactness checked on stalks (Etale Cohomology, Theorem etale-cohomology-theorem-exactness-stalks, canonical tag 03PU). cited at proof lines 994-996. Right exactness of f_! may be checked on stalks, reducing part (2) to part (1).
- Fibres of locally quasi-finite morphisms (Morphisms, Lemma morphisms-lemma-locally-quasi-finite-fibres, canonical tag 06RT). cited at proof lines 1001-1003. The fibre over a geometric point has discrete underlying topological space.
- Stalks of f_! for separated morphisms (Lemma lemma-stalk-f-shriek-separated, canonical tag 0F76). cited at proof lines 1008-1009. The more general stalk computation the formula follows from.

Source reference. Original theorem 03PU: sources/stacks-chow/tags.txt, source lines 4871–4871. Cited at proof lines 994-996: right exactness of f_! may be checked on stalks, reducing part (2) to part (1).

Source reference. Original lemma 06RT: sources/stacks-chow/tags.txt, source lines 8652–8652. Cited at proof lines 1001-1003: the fibre over a geometric point has discrete underlying topological space.

Source reference. Original lemma 0F76: sources/stacks-chow/tags.txt, source lines 18996–18996. Cited at proof lines 1008-1009: the more general stalk computation from which the formula follows.

Lemma 0.2. *Let $f : X \to Y$ be a morphism of schemes which is separated and locally quasi-finite. Then*

(1) *for $\mathcal{F}$ in $Ab(X_{étale})$ and a geometric point $\overline{y} : \mathrm{Spec}(k) \to Y$ we have*

$$(f_!\mathcal{F})_{\overline{y}} = \bigoplus\nolimits_{f(\overline{x})=\overline{y}} \mathcal{F}_{\overline{x}}$$

functorially in $\mathcal{F}$, and

(2) *the functor $f_!$ is exact.*

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Proof. The functor $f_!$ is left exact by construction. Right exactness may be checked on stalks (Étale Cohomology, Theorem 03PU). Thus it suffices to prove part (1).

Let $\overline{y} : \mathrm{Spec}(k) \to Y$ be a geometric point. The scheme $X_{\overline{y}}$ has a discrete underlying topological space (Morphisms, Lemma 06RT) and all the residue fields at the points are equal to k (as finite extensions of k). Hence $\{\overline{x} : \mathrm{Spec}(k) \to X : f(\overline{x}) = \overline{y}\}$ is equal to the set of points of $X_{\overline{y}}$. Thus the computation of the stalk follows from the more general Lemma 0F76. □